



Comparative abiotic, regulatory-aligned assessment of nutrient and trace-metal release from secondary geomaterials for soil application

G. Galamini¹ · G. Malvolti¹ · M. Lezzerini² · D. Malferrari^{1,3} · A. F. Gualtieri^{1,4}

Received: 4 May 2026 / Revised: 23 July 2026 / Accepted: 16 August 2026
© The Author(s) 2026

Abstract

This study provides an abiotic, regulatory-aligned leaching assessment of two secondary geomaterials intended for soil application: RE.WO, a secondary raw material derived from the thermal transformation of mineral wool construction waste, and BO50, a by-product of rock-cutting operations. The objective is to compare nutrient release and trace-metal mobility as a function of their contrasting compositions. To this aim, mineralogical and bulk-chemical characterization was combined with abiotic, time-resolved, regulatory-aligned *in vitro* leaching tests performed under weakly acidic, chelating, and weakly alkaline conditions. RE.WO is characterized by a fully amorphous glassy matrix that exhibits rapid dissolution kinetics and high cumulative cation release, particularly under chelating conditions. While this behaviour indicates higher short-term extractability of micronutrient elements (notably Mn, Cu, and Zn), it also facilitates the mobilization of regulated trace metals. Conversely, BO50 is dominated by crystalline silicates and K-bearing mineral phases; this refractory nature results in low solubility, gradual and sustained release of potassium and comparatively low mobilization of regulated trace elements under the tested conditions. These findings support a risk-informed approach to agronomic utilization. In terms of regulatory positioning under European Regulation 2019/1009, BO50 appears more readily compatible with possible classification as a virgin material or by-product, although conditional on verification of by-product status, registration under applicable chemicals legislation, and compliance with contaminant requirements. Conversely, the higher mobilization of regulated trace elements in RE.WO may preclude its current classification under European Regulation 2019/1009 pathways and its use appears more defensible in restricted non-food applications.

Keywords Agronomic valorisation · Basalt · Component Material Categories · End-of-Waste · European Regulation 2019/1009 · Mineral wool · Soil improvers

Editorial responsibility: Samareh Mirkia.

✉ D. Malferrari
daniele.malferrari@unimore.it

¹ Department of Chemical and Geological Sciences, University of Modena and Reggio Emilia, Via G. Campi 103, 41125 Modena, Italy

² Department of Earth Sciences, University of Pisa, Via Santa Maria 53, 56126 Pisa, Italy

³ BIOGEST-SITEIA, Piazzale Europa 1a, 42124 Reggio Emilia, Italy

⁴ Inter-Departmental Research and Innovation Centre On Construction and Environmental Services of the University of Modena and Reggio Emilia, Via P. Vivarelli 10, 41125 Modena, Italy

Introduction

Today, agriculture must address the growing need to increase production in a sustainable way while reducing dependence on traditional and synthetic fertilizers, whose production and use are associated with significant environmental impacts (Nakachew et al. 2024; Quille et al. 2025; Sharma and Sharma 2025). Among several emerging options, rock powders have shown potential in two main distinct application contexts: (i) as particle film for foliar use against biotic and abiotic stresses (Glenn and Puterka 2004; Sharma et al. 2015; Salerno et al. 2021; Rotondi et al. 2022; Petoumenou 2023; Galamini et al. 2024b); (ii) as mineral amendments when directly incorporated into the soil to influence its physical and chemical properties (Faccini et al. 2015; Eslami et al. 2018; Datta et al. 2020; Cataldo et al. 2021;



Galamini et al. 2024a; Reddy et al. 2024). Regarding the latter, however, larger particle sizes are sometimes preferable for soil applications compared to the finer fractions typically found in powders (e.g., De Smedt et al. 2015; Mi et al. 2020; Malferrari et al. 2021; Jarosz et al. 2022; Aslan and Arslan 2024).

The most widely known and used types of rock powder can be grouped into three main categories, namely rock powder rich in kaolinite (frequently cited as kaolin, i.e., the sedimentary rock composed mainly of the mineral kaolinite), smectite (i.e., clays, bentonites) and zeolites (i.e., zeolites, zeolitized/zeolitic tuffs). There is extensive literature on the efficiency of these natural and safe geomaterials (raw or modified in various ways) as well as on their methods and fields of application (Borah et al. 2022; Chen et al. 2024; Chokri et al. 2025; Sarkar et al. 2025; Goyal et al. 2025). However, various alternative rock powders are increasingly appearing on the market: a quick search using keywords such as “powder”, “rock” and “agriculture” on common search engines yields numerous commercial advertisements for several types of mineral-based products. Nevertheless, compared with well-established geomaterials such as kaolin, bentonites and zeolitic tuffs, these “emerging” rock powders are supported by a more limited body of agronomic evidence and therefore require careful case-by-case evaluation. Among these products, basalt powder, and powders derived from rock with similar chemical and mineralogical composition, represent one of the most prominent and rapidly expanding categories.

Basalt is an igneous volcanic rock, formed by the rapid cooling of basaltic lava at or near the Earth’s surface or in water. It is a mafic rock, typically containing glass and phenocrysts of plagioclase (mostly anorthite), pyroxenes and sometimes olivine. Compared with felsic rocks, basalt generally has lower silica content (typically about 45–52 wt. %) and lower alkali contents (2–6 wt. %, except in alkaline varieties), while being relatively enriched in Fe, Mg and Ca (5–12 wt. %).

In agriculture, basalt is increasingly recognized as a multifunctional soil improver due to its capacity for the sustained release of nutrients and its ability to ameliorate a wide range of soil chemical and physical properties (Anda et al. 2015; Hartatik et al. 2019; Swoboda et al. 2022; Benevides Filho et al. 2023; Son et al. 2024; Rodrigues et al. 2024, 2025). Basalt application enhances soil fertility by increasing cation exchange capacity and base availability, which improves nutrient retention and reduces leaching, particularly in highly weathered soils such as Oxisols (Abdul Halim et al. 2018; Conceição et al. 2022; Benevides Filho et al. 2023). The relatively slow dissolution of basalt ensures a sustained release of essential macro- and micronutrients, while simultaneously buffering pH in soils where natural acidity and nutrient depletion constrain crop productivity

(Nunes et al. 2014; Abdul Halim et al. 2018; Hartatik et al. 2019; Chigilino et al. 2022; Korchagin et al. 2022; Conceição et al. 2022; Luchese et al. 2023b). Furthermore, soluble silicon species released during weathering may enhance plant resistance to both biotic and abiotic stresses (De Carvalho Ribeiro et al. 2025). Beyond enhancing base cation availability, basalt application influences soil phosphorus dynamics by facilitating its desorption from soil colloids and mitigating its fixation by iron and aluminium oxides, thereby increasing the bioavailability of this critical nutrient (Korchagin et al. 2022; Luchese et al. 2023a). However, soil-specific characteristics can lead to adverse effects or variable outcomes that warrant further investigation (Desmalles et al. 2025).

Basalt addition also improves soil aggregation and porosity, enhancing water retention and aeration, which are critical for plant root growth, microbial activity and resilience in sandy or degraded soils (Sah et al. 2022; Dupla et al. 2024; De Carvalho Ribeiro et al. 2024). Furthermore, recent studies highlighted its role in carbon sequestration: in fact, as basalt weathers, it consumes atmospheric and soil CO₂, leading to the formation of stable bicarbonate and carbonate species contributing, over time, to long-term carbon storage (Lewis et al. 2021; Swoboda et al. 2022; Desmalles et al. 2025).

Beyond these benefits, the application of rock powders involves context-dependent limitations that require the careful evaluation of factors such as application rate, particle size, soil type, and nutrient release kinetics (Swoboda et al. 2022; Desmalles et al. 2025). Additional practical constraints include the requirement for large volumes of finely ground material, which entails significant grinding and transport costs. To address these challenges, recent research has focused on the valorisation of basalt by-products from various industrial streams, effectively integrating their application into a circular economy framework (Nunes et al. 2014; Dalmora et al. 2016; Korchagin et al. 2019; Luchese et al. 2023b).

An even more promising approach involves the use of secondary raw materials, derived from the thermal treatment of hazardous waste, with chemical (macro- and micronutrient content) and physical (shape, size, and hardness of grains) properties comparable to those of basalt. These include RE.WO, an innovative material produced using rock and glass wool waste recovered from building demolition sites (Gualtieri and Zanatto 2022; Arletti et al. 2023). Because rock wool is manufactured from mafic source rocks (e.g., basalt and gabbro), RE.WO is chemically analogous to basalt, although some variations in alkali content may occur due to differences in the original raw materials and production processes.

RE.WO is a recent material, and no studies have yet investigated its application (or that of analogous materials)



in agriculture. Therefore, the aim of this research is to evaluate whether RE.WO can be used as a substitute for, or in addition to, basalt powder as a soil improver. To align with a circular economy framework, a powder derived from mafic rock-cutting operations, hereafter designated as BO50 and classified by the suppliers as basalt, was used as a reference material. Previous studies (e.g., Sungur et al. 2015; Botsou et al. 2016; Andrade et al. 2025) have investigated the leaching behaviour and environmental implications of mineral and waste-derived materials under controlled conditions, highlighting the importance of both bulk composition and extractive environment in controlling element mobility. In line with these approaches, the present study combines mineralogical and chemical characterization with abiotic, time-resolved, regulatory-aligned *in vitro* leaching tests conducted under weakly acidic, chelating and weakly alkaline conditions to compare nutrient release and the mobilization of selected potentially toxic metals. This framework supports the comparative assessment and regulatory positioning of the two materials. As these materials lack nitrogen and their phosphorus content is too low to justify a specific investigation as a significant nutrient source, their agronomic value should primarily be interpreted in terms of potassium supply (particularly for BO50), micronutrient release, and soil-improving potential, rather than as complete fertilizers.

This research was carried out in 2025/2026 in northern Italy, using secondary geomaterials derived from Italian industrial streams and characterized under controlled laboratory conditions.

Materials and methods

Materials

The materials investigated were RE.WO, a secondary raw material developed within an End-of-Waste valorisation framework through the thermal treatment of rock and glass wool sourced from construction and demolition waste (Gualtieri and Zanatto 2022; Arletti et al. 2023), and BO50, a residual powder generated as a by-product during the industrial cutting and shaping of basaltic rocks. This latter material comprises a mixture of basalt and petrographically related lithotypes with a similar chemical composition.

Both materials were selected as representative secondary mineral resources of potential interest for soil-related applications, with RE.WO also reflecting the valorisation potential of an abundant mineral wool waste stream in Europe (Väntsi and Kärki 2014). Despite its recent development and limited production history, the analysed RE.WO batches showed relatively low variability in bulk chemical composition, particularly for major elements. This preliminary consistency is plausibly related to the use

of large and relatively homogeneous batches of glass and rock wool derived from similar source rocks. However, longer-term production monitoring is required to confirm compositional stability over time, especially for regulated trace elements (see Sect. “[Chemical and mineralogical characterization](#)”).

Materials characterization

An analytical workflow combining X-ray powder diffraction (XRPD), particle-size analysis, X-ray fluorescence (XRF), ICP-based solution chemistry and scanning electron microscopy (SEM) was applied to characterize the materials and the eluates obtained from the leaching tests. XRF was employed for the analysis of solid samples, whereas Inductively Coupled Plasma Mass Spectrometry (ICP-MS) and Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) were used to investigate the solutions obtained from the leaching tests (see Sect. “[Leaching tests](#)”). Particle-size distribution was determined by laser diffraction following ISO 13320. Full instrumental settings, details on the methods, instrumentation, and experimental conditions are provided as Supplementary Information (SI-1).

More specifically, ICP-OES was used to determine Na, Mg, K, Ca, and Fe, whereas ICP-MS was employed for V, Cr, Mn, Co, Ni, Cu, Zn, As, Mo, Cd, Sb and Pb. Among these, Mn, Cu, and Zn are considered essential micronutrients, whereas the others, although in some cases depending on the amount and product category (Ni, Co, Mo) and/or oxidation state (Cr), are potential undesirable pollutants as indicated in the European and Italian Fertilizing Products Regulation (DGL75-2010, 2010; EU 2019/1009, 2019). Therefore, for discussion purposes, trace elements are considered in relation to their main relevance within the present study, namely agronomic interest (e.g., Cu and Zn) or regulatory concern. This operational grouping does not imply that some elements (e.g., Ni or Mo) with known plant essentiality are excluded a priori from micronutrient status; rather, in the present work they are discussed conservatively because of their regulatory significance and concentration-dependent risk profile. Moreover, for consistency with the analytical framework adopted here, the investigated elements are discussed as major constituents (Na, Mg, K, Ca, Mn and Fe) and trace elements (V, Cr, Mn, Co, Ni, Cu, Zn, As, Mo, Cd, Sb and Pb). This grouping is operational and reflects their abundance in the studied materials and their analytical treatment; it does not imply a plant-nutritional classification of all major constituents as macronutrients (for example, from a plant-nutritional perspective, Fe, and Mn, like Cu and Zn, are classified micronutrients).



Leaching tests

The methodological framework adopted herein follows the protocols recommended in previous studies (e.g., Ramos et al. 2015, 2017), with adaptations implemented to ensure alignment with European and Italian Fertilizing Products Regulation (DGL75-2010, 2010, EU 2019/1009, 2019). However, nutrient mobilization significantly depends on soil properties and biological activity, while root and rhizosphere processes, microbial metabolites, redox gradients, and mineral transformations dynamically regulate element mobility through competing dissolution and immobilization mechanisms (Jones 1998; Violante et al. 2010). Consequently, the tests performed in this study should be interpreted as defining an abiotic extractability potential, rather than as quantifying biological availability or plant uptake. On this basis, and to provide a reproducible comparison between the two materials, abiotic *in vitro* leaching experiments were carried out on both RE.WO and BO50 samples in three different environments (Khademi et al. 2009; Ding et al. 2014; Sur et al. 2023): weakly acidic (2% v/v acetic acid), chelating (2% v/v citric acid), and weakly alkaline (pH 8.0 aqueous solution obtained by adding a few drops of ammonia to Millipore water). Acetic acid is a weak monocarboxylic acid, typical of many organic soil solutions (e.g., humic and fulvic acids), and simulates the acidic environment of soils rich in humic substances. Citric acid has a more pronounced chelating function (triple carboxylic) and may simulate active rhizospheric environments where roots and microorganisms release organic acids to mobilize nutrients. The pH 8.0 solution simulates carbonate or arid soils where precipitation or adsorption processes prevail over dissolution.

To evaluate the kinetics of element release, tests were performed at various contact times. Because BO50 has a finer grain size than RE.WO, the latter was ground prior to testing to obtain a comparable grain-size distribution (Fig. 1). Leaching experiments were conducted under continuous orbital agitation using an IKA KS 130 basic shaker (400 rpm). All the tests were carried out in capped polypropylene tubes at room temperature (20 °C), with a solid-to-solution ratio of 1:20 (0.5 g of sample in 10 mL of solution). To investigate release kinetics, tests were terminated after 0.25, 0.5, 1, 2, 3, 4, 8, 12, 24, 36, 48, and 72h. At each time point, suspensions were filtered through 0.22 µm SIR nylon filters. Eluates were then acidified by adding 100 µL of 3% v/v HNO₃ to stabilize the solutions. Dilution factors were applied to calculate the effective concentrations relative to the initial sample mass. Blank controls (10 mL of 2% v/v citric acid, 2% v/v acetic acid, and alkaline deionized water) were processed in parallel under identical conditions.

Leaching experiments were performed in triplicate; at each sampling time, independent tubes were sacrificed for analysis, and results are reported as the mean of these

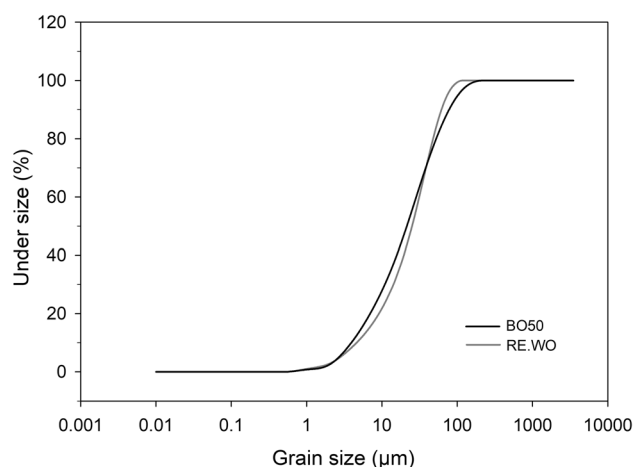


Fig. 1 Particle size distribution of BO50 and RE.WO, the latter after grinding to normalize particle size distribution of the two geomaterials

replicates. pH was not monitored during the experiments because the experimental design was intended to compare the operational short-term extractability of the two materials under fixed and standardized extractive conditions, in line with the regulatory-oriented framework adopted in this study, rather than to reproduce the progressive evolution of soil pore-water chemistry. Accordingly, potential buffering effects of the materials were not assessed and should be considered when interpreting the results and extrapolating them to real soil systems. The 72h duration was selected to resolve early-stage release kinetics within the standardized, regulatory-aligned extractability framework adopted in this study. Consequently, these experiments should not be interpreted as simulations of long-term field behaviour. Instead, longer-term experiments (incubations, columns, and plant growth) are required to evaluate nutrient release and trace-metal mobility under realistic conditions, as outcomes will heavily depend on specific soil properties.

Statistical analysis

To provide a simple statistical comparison between BO50 and RE.WO under each extractive condition, cumulative release values at the 72h endpoint were analysed separately for major constituents and trace elements. The 72h endpoint was selected to avoid pseudo-replication associated with the use of cumulative time-series data and to provide a common summary of the final operational release under each condition. For each extractant, BO50 and RE.WO were compared using paired t-tests on log₁₀-transformed cumulative release values, with elements treated as paired observations. The tested variable was therefore log₁₀(RE.WO/BO50), where a value of zero indicates no systematic difference between the two materials. Non-detected values were excluded from



formal paired comparisons, and the number of matched elements is reported for each test. *P*-values were adjusted for multiple comparisons using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

Results and discussion

Particle size characterization

The particle size distributions of BO50 and RE.WO are shown in Fig. 1. As supplied, RE.WO is slightly coarser than BO50, although its pristine grain size is still compatible with direct practical use without further grinding. However, because particle size can significantly influence dissolution kinetics and element release, RE.WO was ground exclusively for the leaching experiments in order to match, as closely as possible, the granulometric profile of BO50 and thus enable a robust and meaningful comparison of leaching behaviour. After this standardization step, both materials displayed comparable particle size distributions: 87.6% of RE.WO and 82.4% of BO50 particles fell within the 2–63 µm range, whereas the <2 µm fraction accounted for 2.8% and 2.5%, respectively. These comparable granulometric profiles support the interpretation of the observed differences in leaching behaviour as being primarily controlled by mineralogical and chemical factors rather than by particle-size effects.

Chemical and mineralogical characterization

The bulk chemical and mineralogical compositions determined by XRF and quantitative phase analysis (QPA) are reported in Table 1. The low standard deviations observed for RE.WO across different batches indicate a substantially stable composition for the material. However, since RE.WO is a recently developed material, this stability requires further confirmation. The variation in BO50 is even more restricted because the measurements were conducted on different batches but originating from the same sources (same quarry by-product) and XRF is an analytical method characterized by high accuracy and precision.

Although BO50 is designated as basalt based on the geological context of its source rock, chemical and mineralogical analyses classify the powder as a leucite-bearing phonotephrite/tephri-phonolite. Nevertheless, as noted above, this distinction does not affect the comparison with RE.WO or its potential application in agriculture. In fact, BO50 is enriched in potassium, reflecting its high content of K-bearing phases such as leucite and K-feldspar (orthoclase and sanidine), making it a promising source of exchangeable potassium for soils. In contrast, BO50 contains less Mg, Mn and particularly Ca and Fe. Among the trace elements (Tables 2 and

Table 1 XRF measurement (reported in conventional oxide wt. %) and QPA (phase wt. %) of BO50 and RE.WO

	BO50	RE.WO		BO50
Na ₂ O	2.09 (0.12)	5.56 (0.41)	Amphibole (hbl)	1.1 (6)
MgO	2.30 (0.16)	5.12 (0.59)	Biotite (phl)	2.0 (4)
Al ₂ O ₃	19.8 (0.8)	7.04 (0.28)	Hematite	1.2 (7)
SiO ₂	50.0 (1.1)	47.8 (2.16)	K-feldspar (or)	5.8 (1)
P ₂ O ₅	0.42 (0.04)	0.18 (0.09)	Leucite	29.6 (3)
K ₂ O	8.31 (0.66)	0.86 (0.11)	Pyroxene (aug)	20.8 (3)
CaO	7.91 (0.51)	19.3 (1.16)	Plagioclase (an)	13.1 (2)
TiO ₂	0.71 (0.08)	0.78 (0.28)	Sanidine	8.4 (2)
MnO	0.14 (0.02)	0.42 (0.08)	Amorphous	18.0 (6)
Fe ₂ O ₃	7.45 (0.77)	12.0 (0.96)	χ ²	5.549
LOI	0.40 (0.08)	0.11 (0.08)	R _p (%)	5.67
			R _{wp} (%)	7.77

LOI is the Loss On Ignition (weight loss % at 1100 °C); totals sum to 100% when including all trace elements (Table S1). The QPA of RE.WO is not reported because the material is composed solely of amorphous glass. In XRF measurements values in parenthesis are the standard deviation among three different batches, while in QPA they denote error calculated through GSAS software (see SI-1). The literature reference for the structural models used in QPA are: amphibole (hornblende structure) (Mancini et al. 1996), pyroxene (augite structure) (Gualtieri 2000), biotite (phlogopite structure) (Brigatti et al. 2008), hematite (Blake et al. 1966), leucite (Palmer et al. 1997), plagioclase (anorthite structure) (Harlow 1982), quartz (Le Page and Donnay 1976), sanidine (Yamnova et al. 2000)

S1), Cu and Zn occur at appreciable concentrations, with RE.WO showing higher values than BO50.

These data indicate that both materials contain measurable amounts of potentially agronomically relevant micro-nutrients; however, total concentrations alone do not necessarily reflect their actual bioavailability for plant uptake. At the same time, both materials also contain non-negligible concentrations of regulated trace metals (Table 2), which may have implications for their environmental compatibility and regulatory classification.

Leaching tests for major constituents

The cumulative release (mg/Kg) of Na, K, Ca, Mg, Fe and Mn under the three extraction conditions is illustrated in Figs. 2, 3 and 4. Data normalized to the bulk content (i.e., the percentage released relative to the total elemental content reported in Table 1) are presented in Figures S1, S2 and S3. All numerical values corresponding to the graphs discussed in this section are reported in Appendix-A.

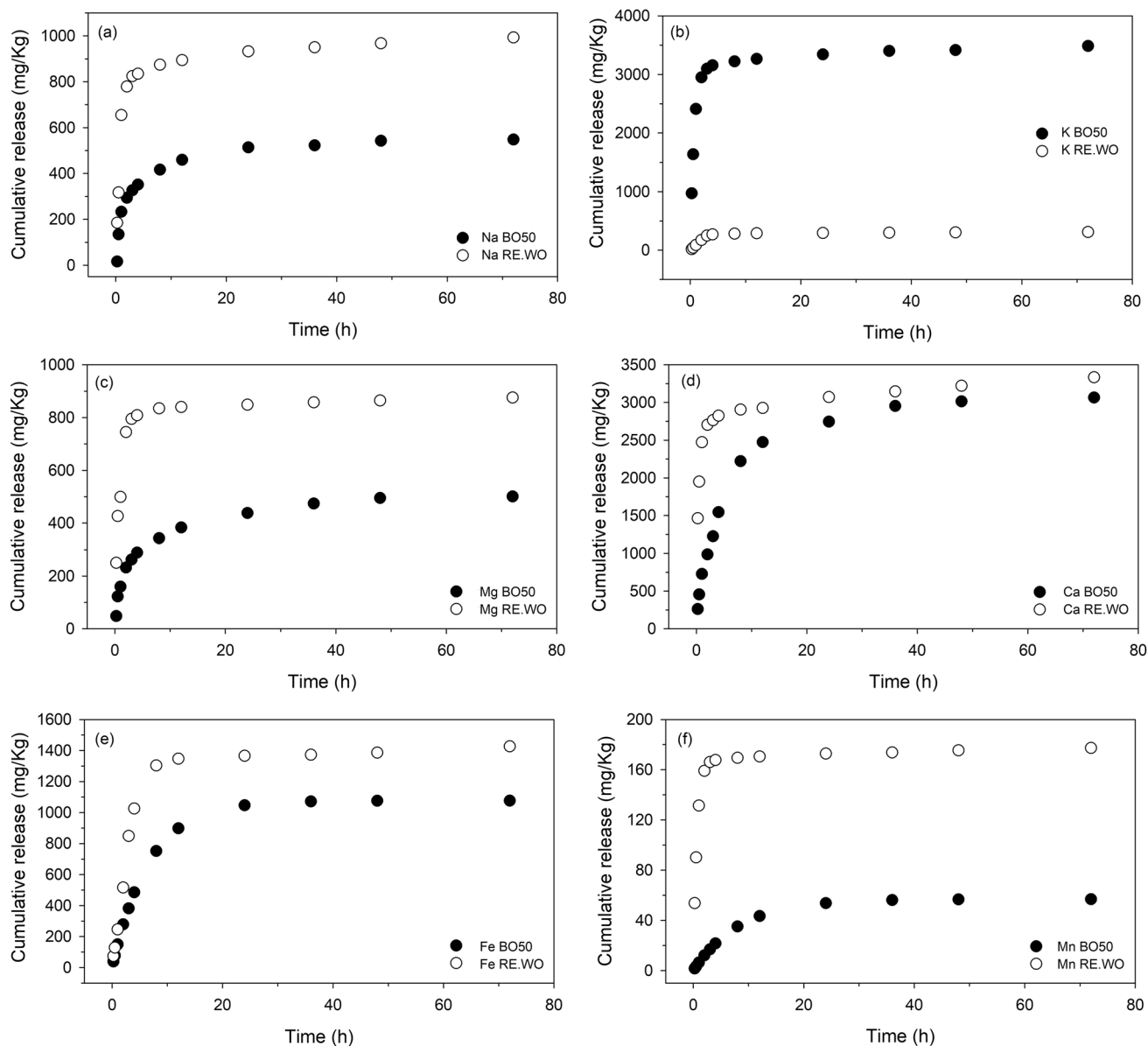
Across all conditions, cumulative release increased sharply during the initial hours of contact, most notably within the first 8h, followed by a gradual deceleration in the dissolution rate until an apparent equilibrium plateau was reached after approximately 36–48h. Nevertheless,



Table 2 Selected trace elements concentration (mg/Kg) from the same batches used for major elements (major and trace elements are measured on the same samples)

	Cu	Zn	V	Cr	Co	Ni	As	Mo	Cd	Sb	Pb
BO50	11 (1)	78 (8)	217 (11)	21 (3)	21 (4)	11 (2)	12 (4)	4 (1)	5 (2)	0.28 (0.09)	68 (9)
RE.WO	279 (36)	1145 (98)	128 (31)	709 (86)	28 (9)	229 (38)	3 (1)	19 (6)	12 (5)	12 (4)	32 (9)

The concentration of Sb in BO50 was measured on mineralized sample using ICP-MS being below detection limit in XRF. Values in parenthesis are standard deviation among three replicates. See SI-1 for additional details on the method and Table S1 for the full data-set of trace elements

**Fig. 2** Leaching of major elements using 2% acetic acid as extractant. The specific element is indicated in the legend of each panel. Error bars are not shown as they fall within the symbol size

the leaching behaviour revealed distinct dissolution patterns for the two materials, governed primarily by their contrasting mineralogical composition and the solubility of

constituent phases (Table 1). The higher dissolution rates and total released concentrations observed for RE.WO (with some exceptions discussed below) across all conditions are



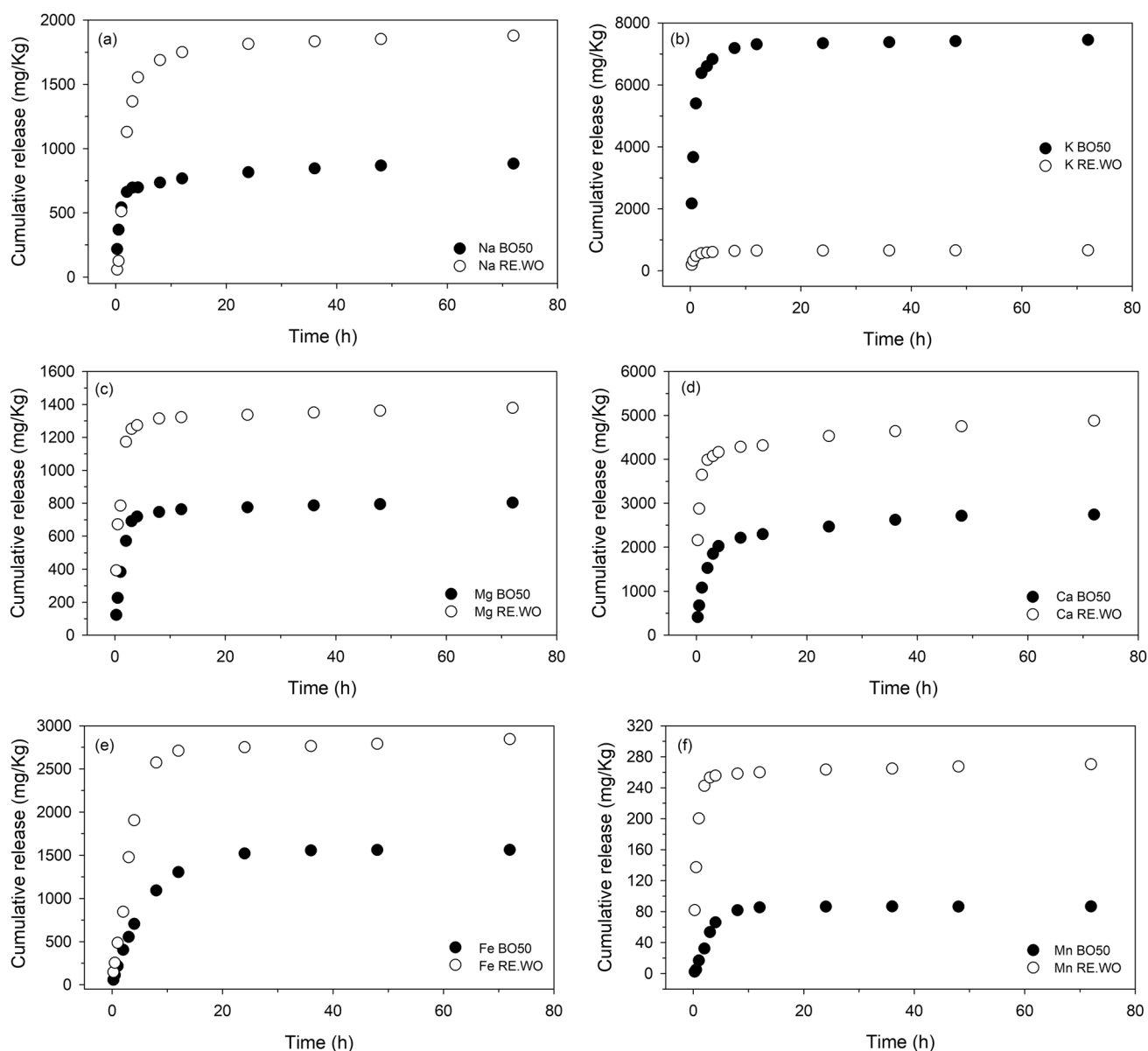


Fig. 3 Leaching of major elements using citric acid 2% as extractant. Details as in caption of Fig. 2

consistent with its amorphous, glass-dominated nature characterized by a mechanically fragile structure and conchoidal fracture surfaces (Fig. S4). In comparison, BO50, which is predominantly composed of crystalline silicates (amorphous content: 18.0 wt. %, Table 1), exhibited markedly lower solubility and slower, or at most comparable, release kinetics.

Under weakly acidic conditions (Fig. 2), both materials exhibited moderate solubility, but the extent and selectivity of dissolution varied according to chemical and mineralogical composition, with BO50, except for K, showing generally slower release kinetics. RE.WO, consistently with its chemical composition, released larger proportions of Na, Mg, Ca, and Fe (Figs. 2a, c–e), with Na, Mg and Ca showing

the steepest initial slopes during the first 4h. The cumulative release of Fe (Fig. 2e) also increased notably, but tended to stabilize a little later. BO50 displayed lower cumulative release of these elements throughout the test (Figs. 2a, c–e), which is consistent with its minor bulk content (Table 1). However, with the (minor) exception of Ca, it showed comparable normalized release relative to RE.WO (Fig. S1). The normalized release of Mn was comparable despite lower total concentration in BO50. Interestingly, although BO50 is markedly enriched in K (8.31 wt. % K_2O versus 0.86 wt. % in RE.WO, Table 1), its K release was lower than might be expected (Fig. 2b), and the normalized release (Fig. S1) is comparable to that of RE.WO. This difference is due to



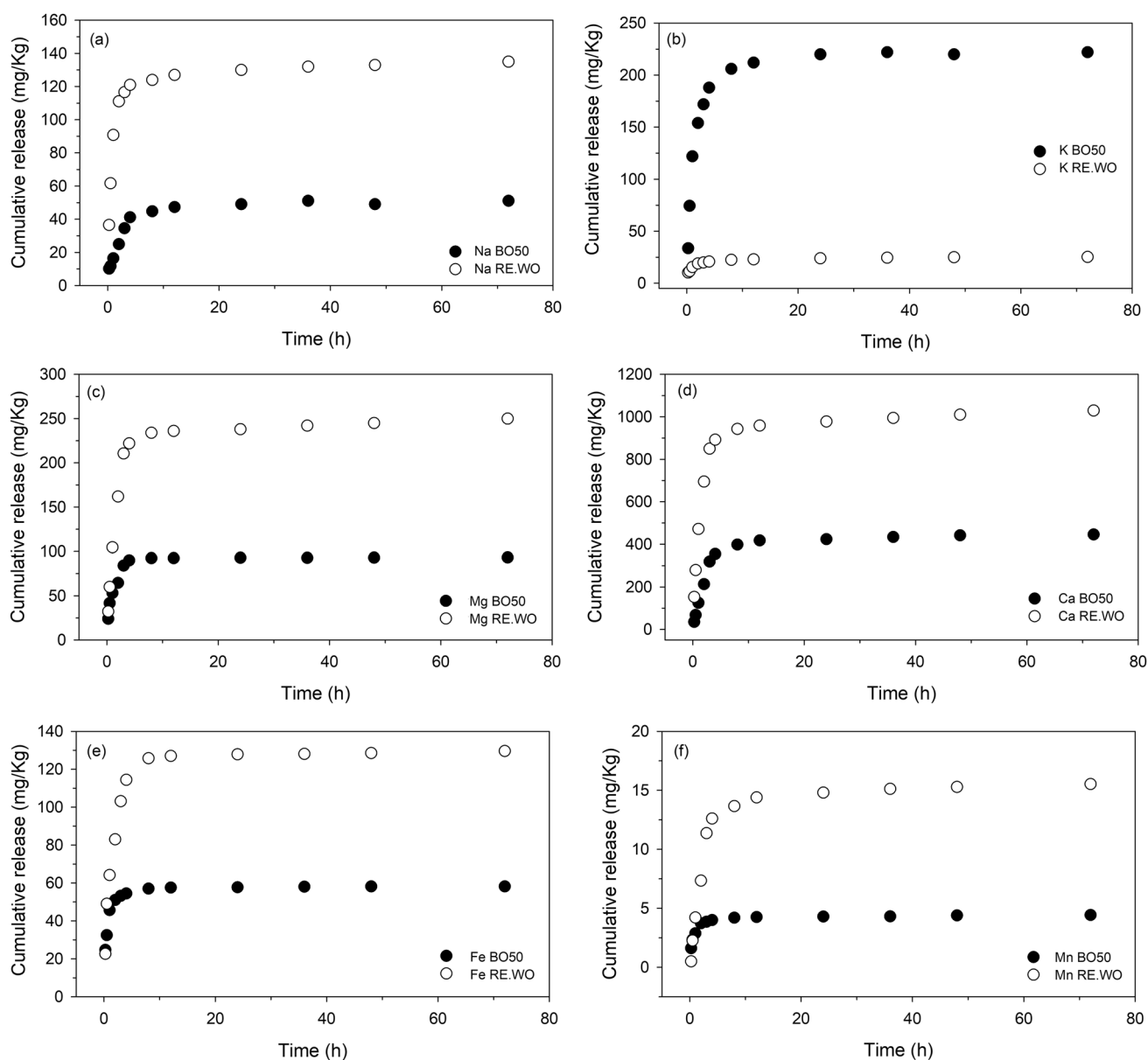


Fig. 4 Leaching of major elements using water pH 8.0 as extractant. Details as in caption of Fig. 2

mineralogical controls: in BO50, K is structurally bound within leucite and K-feldspar frameworks, both characterized by low solubility, while in RE.WO K occurs in more labile, glassy forms, promoting greater release despite lower bulk content.

In citric acid (Fig. 3), dissolution was markedly enhanced compared with acetic acid for both materials. The cumulative release curves (Figs. 3a–f) showed pronounced increases for nearly all major cations. The cumulative release of Na and Mg (Fig. 3a, c) showed substantial increases compared with acetic acid extraction; similarly, the chelating capacity of citrate enhanced the mobilization of Fe and Mn (Fig. 3e, f). In contrast, Ca in BO50 exhibited a high but rapidly

stabilized release profile (Fig. 3d). While the cumulative release of K was again dominant (Fig. 3b), its normalized release (Fig. S2) remained consistent with levels observed during acetic acid extraction (Fig. S1), further confirming that elemental mobility is primarily controlled by mineralogical factors rather than bulk concentration.

In RE.WO, Fe release approached the plateau after 48h (Fig. 3e), suggesting sustained complexation and solubilization despite its higher bulk concentration relative to BO50 (Table 1). Fe release from BO50 was slower than in RE.WO, but increased progressively with time, reflecting partial dissolution of Fe-bearing silicates. Overall, citric acid induced the highest dissolution efficiency for both samples, with



a distinct element-selective behaviour reflecting the differences between the glassy (RE.WO) and predominantly crystalline (BO50) matrices.

Under weakly alkaline conditions (Fig. 4), total releases were significantly lower for all elements (Fig. 4a–f). Both materials exhibited nearly flat concentration profiles, approaching equilibrium after about 8h, which indicates extremely limited solubility. Only Ca and Mg showed a release greater than 0.5% of the initial content (Fig. S3), while the other elements remained largely immobile under these conditions.

Leaching tests for trace elements

For trace elements, both cumulative (Figs. 5, 6 and 7, panels a and b) and normalized (Figs. 5, 6 and 7, panels c and d) release are presented in the main text, as their combined interpretation is necessary to distinguish between concentration-driven effects and mineralogical or matrix-related controls on element mobility. Given the wide range of concentrations among the analysed elements, logarithmic scales are used to improve data readability and to allow meaningful comparison across elements occurring at different concentration levels. All numerical values corresponding to the graphs discussed in this section are reported in Appendix-A.

RE.WO generally exhibits higher cumulative release and faster kinetics than BO50, particularly in organic acid media, with V, As, and Pb being the primary exceptions. Dissolution kinetics in both materials show characteristic trends: a steep initial increase in the first few hours, followed by a progressive shift toward a quasi-plateau between 36 and 72h. This quasi-plateau is more pronounced in BO50, whereas RE.WO tends to exhibit multi-stage (two-step) release patterns for several elements (as detailed below). However, while the magnitude and selectivity of release of major elements appear to be predominantly controlled by the nature of the matrix (crystalline versus glassy), for trace elements both the bulk composition (Table 2) and matrix-related factors play a key role in governing their mobility.

Leaching tests for selected micronutrients (Cu and Zn)

RE.WO exhibits a substantially higher cumulative release of both Cu and Zn, particularly in citric acid, consistent with its greater total metal content. The time-dependent release profiles in acetic and citric acid (Figs. 5a, b and 6a, b, respectively) show a rapid initial increase within the first few hours, followed by a slower, nearly linear rise up to 72h. This behaviour suggests sustained leaching controlled by a combination of glass dissolution and ligand complexation processes. In contrast, BO50, despite its lower cumulative release, displays a more complex normalized release pattern. Under acidic conditions (Figs. 5c, d), its normalized release

remains consistently lower than that of RE.WO. However, in the presence of chelating agents (Figs. 6c, d), BO50 exhibits a consistently higher normalized release than RE.WO. Under alkaline conditions (Figs. 7c, d), the behaviour becomes element-dependent: BO50 shows a lower normalized release than RE.WO only during the initial stages (2h for Cu and 0.5h for Zn).

Considering bulk composition, Zn is the most soluble element in RE.WO. Its release is characterized by a pronounced initial increase followed by a plateau between 8 and 12h. Conversely, Zn release from BO50 is more limited, suggesting that it is primarily controlled by the crystal lattice rather than by its bulk concentration. Citric acid (Fig. 6) significantly enhances the mobilization of both Cu and Zn in both materials, with RE.WO reaching the highest cumulative release values (Figs. 6a, b). The sustained increase up to approximately 36h, together with the consistently high normalized release (exceeding 50% in both materials; Figs. 6c, d) indicates continuous complexation by citrate ligands.

Under alkaline conditions (Fig. 7), both materials exhibit low but measurable trace-element release. Although RE.WO shows higher cumulative release (Figs. 7a, b), differences in normalized release (Figs. 7c, d) are transient and limited to the first few hours of interaction.

Leaching test for selected potentially toxic elements (V, Cr, Co, Ni, As, Mo, Cd, Sb and Pb)

As observed for Cu and Zn, the leaching patterns of potentially hazardous trace elements (Figs. 5, 6 and 7,) show clear differences between RE.WO and BO50.

The normalized release kinetics for BO50 (Figs. 5c, 6c and 7c) are characterised by a short but steep increase during the first 2–4h, followed by a rapid transition to a pronounced plateau, suggesting that leaching is controlled by the rate of transfer from readily accessible surface fractions. This behaviour is consistent across all extraction conditions and largely independent of bulk concentrations. Among these elements, V displays the most evident early-time increase, reaching approximately a normalized release of 20% in acetic acid and 60% in citric acid, but remaining near the detection limit under weakly alkaline conditions.

On the other hand, the normalized release from RE.WO (Figs. 5d, 6d and 7d) typically follows a two-stage trend, with a rapid initial rise during the first 2–8h followed by a slower but continuous increase up to 72h, particularly evident in citric acid. The cumulative release appears primarily controlled by the initial concentration rather than the material matrix. In support of this, Cr, Co, Ni, Mo and Sb exhibit greater mobility in RE.WO than in BO50 under all extraction conditions (Figs. 5a, b, 6a, b and 7a, b). Under citric acid extraction (Figs. 6a, b), the cumulative release of Cd in RE.WO was also greater than from BO50; however, this



Fig. 5 Cumulative (a, b) and normalized (c, d) release of trace elements using 2% acetic acid as extractant for BO50 (a, c) and RE.WO (b, d). Panels (a) and (b) are plotted on a logarithmic scale. Colors correspond to extraction times (see legend in panel a). Missing bars indicate concentrations below the detection limit. Error bars are omitted for graphical clarity

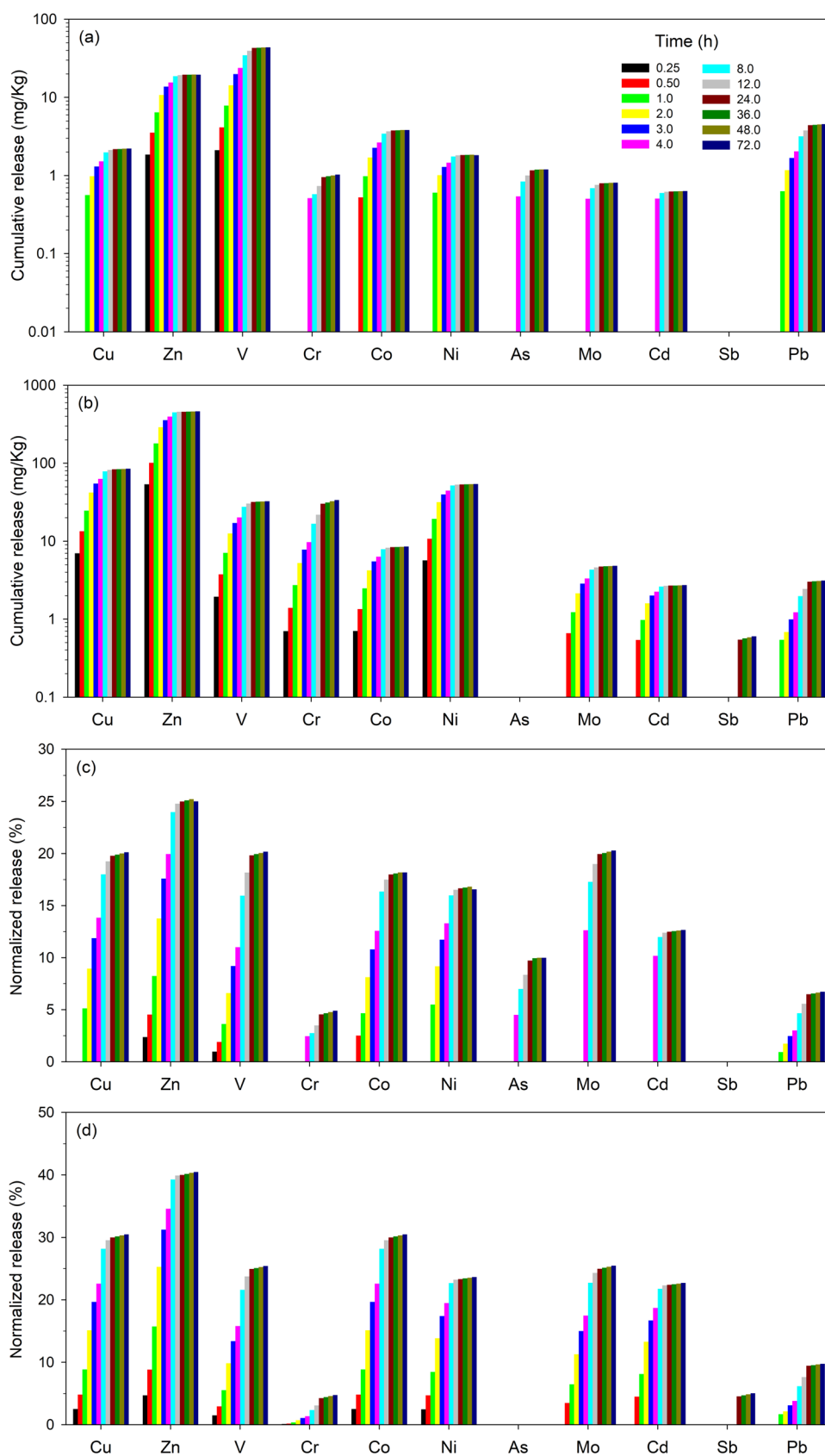


Fig. 6 Cumulative (a, b) and normalized (c, d) release of trace elements using citric acid 2% as extractant for BO50 (a, c) and RE.WO (b, d). Details as in caption of Fig. 5

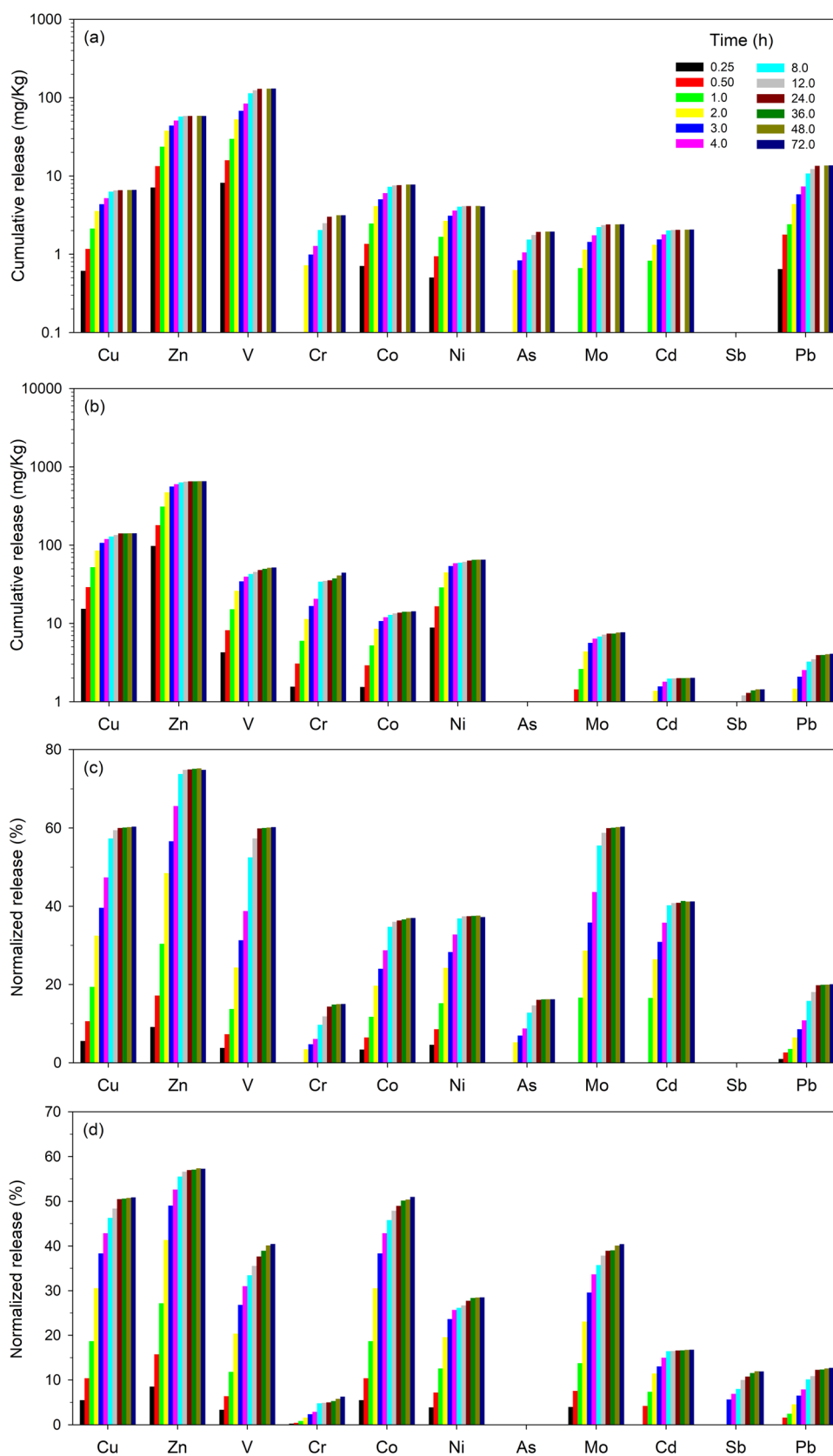
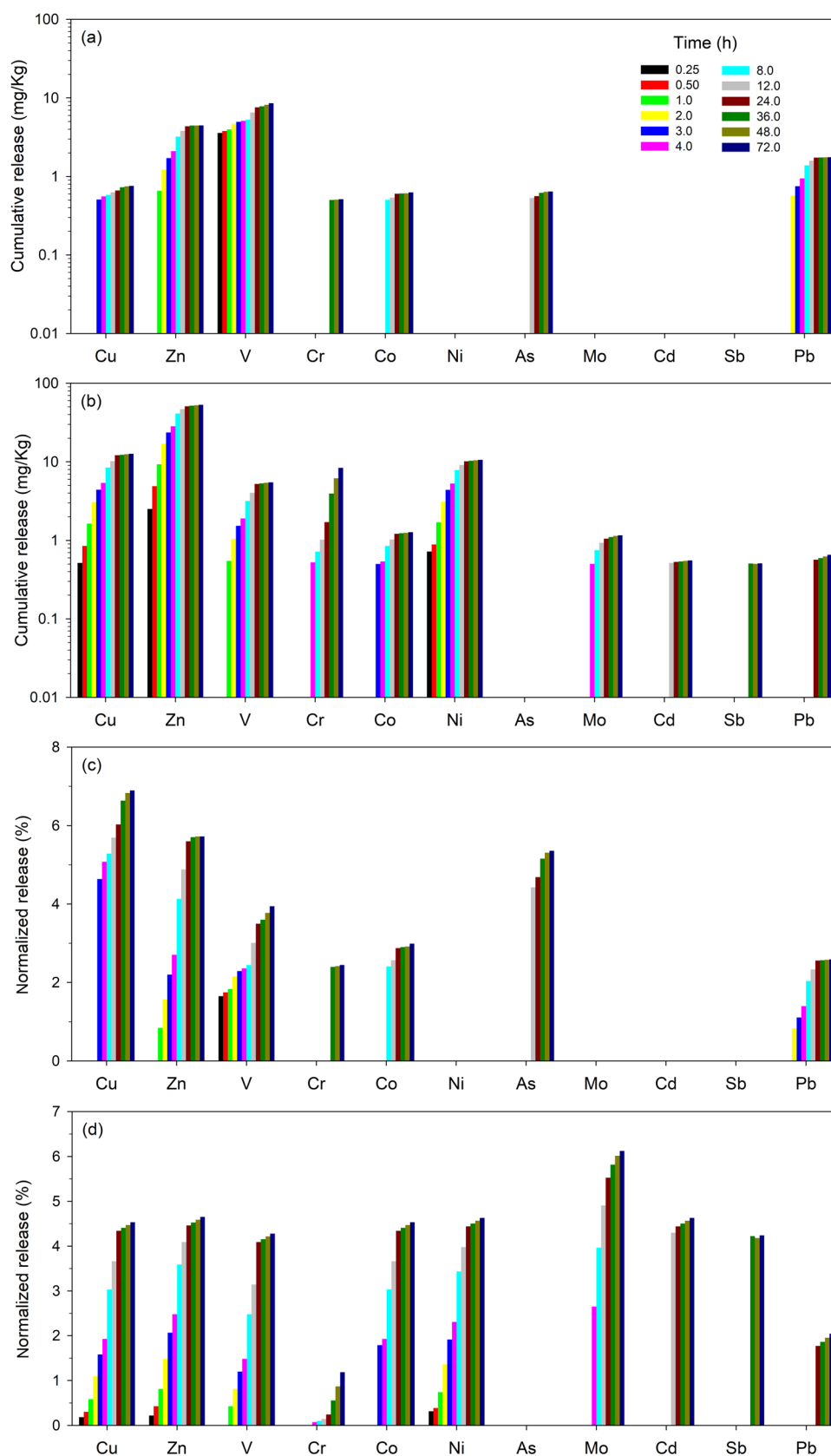


Fig. 7 Cumulative (a, b) and normalized (c, d) release of trace elements using water pH 8.0 as extractant for BO50 (a, c) and RE.WO (b, d). Details as in caption of Fig. 5



difference was only sustained up to 4h of interaction, and the overall cumulative values remained very close up to 72h.

Another specific exception occurs for Mo under chelating conditions: although the cumulative release remains higher from RE.WO, the normalized release only exceeds that of BO50 during the initial 0.5h (Fig. 6c, d). The release pattern of Ni closely mimics that of Cu. Its higher cumulative release from RE.WO is driven by its bulk content, with the greatest mobilization occurring in citric acid and the lowest under alkaline conditions. While the normalized release of Ni from RE.WO is generally greater than from BO50 during acid and alkaline extractions, the relationship reverses in citric acid, where the normalized release is always lower from RE.WO, indicative of strong control by citrate complexation. Cr, in contrast, is the least mobile element in both materials, reflecting its occurrence in less reactive or structurally bound forms, possibly within Fe–Cr oxide phases or the glass matrix. Elemental detection in alkaline solutions was sporadic; in RE.WO, it occurred only after the initial 3h, while in BO50, it was not observed until after 24h.

The cumulative release of V, As and Pb (Figs. 5a, b, 6a, b and 7a, b) is consistently higher from BO50 than from RE.WO, directly correlating with their higher bulk content in BO50 (Table 2). Excluding the previously discussed Cd data, differences in normalized release are primarily confined to V and Pb in acetic acid (Figs. 5c, d). In fact, under acetic acid, the normalized release for both these elements is generally lower from BO50 than from RE.WO, with the exception of Pb during the initial 0.5h (not quantifiable as it falls below the detection limit). For V in alkaline solution (Figs. 7c, d), the normalized release is initially higher from BO50 for the first 4h, but subsequently decreases to values lower than those observed for RE.WO.

Overall, the cumulative release of the analysed elements was low and consistent with their initial bulk content, with the exception of V and, to a lesser extent, Pb. A normalized release exceeding 50% of the initial content was observed for V and Mo (BO50) and Co (RE.WO), exclusively under chelating conditions (Fig. 6). Under acidic conditions

(Fig. 5), the release of these elements is significantly lower, peaking at less than 20% of the bulk content in BO50; the release is further minimized in an alkaline environment (Fig. 7), where all elements showed a release of less than 6% of their bulk content.

From a comparative screening perspective, the regulated trace metals Cd and Pb deserve particular attention. Despite their limited cumulative release (2.7 mg/Kg for Cd from RE.WO using acetic acid; 13.6 mg/Kg for Pb from BO50 using citric acid), their normalized release values are non-negligible (22.6 and 20.1%, respectively). However, the present abiotic leaching data are not intended to provide a quantitative prediction of environmental hazard in soil. Rather, these results suggest that precautionary attention may be warranted for analogous materials containing higher concentrations of these elements, particularly when evaluated against the contaminant thresholds and classification criteria established by the applicable fertilising-product regulations.

The relatively high release of some trace elements compared with major constituents may reflect a combination of processes rather than simple bulk dissolution. Possible contributors include preferential release from surface-accessible fractions, incongruent dissolution of the matrix, stabilization of dissolved trace elements by citrate complexation, and partial re-precipitation of major elements into secondary phases such as carbonates or Fe-rich hydroxides (Lewis et al. 2021; Lu et al. 2024). Although the present dataset does not allow these mechanisms to be resolved quantitatively, they provide plausible geochemical hypotheses for the observed contrast between major- and trace-element release.

Statistical comparison of 72h cumulative release

Statistical comparison of the 72h cumulative release values (Table 3) confirmed that the differences between BO50 and RE.WO were element-group and extractant dependent. For major constituents, no significant global difference between the two materials was detected under any of the three extractive conditions, reflecting the contrasting element-specific

Table 3 Statistical comparison of 72h cumulative release from BO50 and RE.WO

Element group	Extractant	n matched elements	Geometric mean RE.WO/BO50	Median RE.WO/BO50	<i>p</i> -value	BH-adjusted <i>p</i>
ME	Ac. acid 2%	6	1.04	1.54	0.942	0.942
TE	Ac. acid 2%	9	6.66	5.96	0.0078	0.0468
ME	Cit. acid 2%	6	1.22	1.80	0.724	0.868
TE	Cit. acid 2%	9	3.21	3.18	0.066	0.199
ME	pH 8.0 water	6	1.56	2.48	0.436	0.654
TE	pH 8.0 water	6	3.41	6.98	0.138	0.277

Results are reported by extractive condition and element group. *n* indicates matched elements; adjusted *p*-values were calculated using the Benjamini–Hochberg procedure. ME, major elements; TE, trace elements



behaviour, particularly the higher K release from BO50 and the higher release of several other cations from RE.WO. In contrast, at the 72h endpoint, cumulative release across the matched trace-element set was significantly higher from RE.WO than from BO50 under 2% acetic acid extraction, with a geometric mean RE.WO/BO50 ratio of 6.66 and a median ratio of 5.96 (paired t-test on log10-transformed values, BH-adjusted $p = 0.047$). Under citric acid extraction, trace-element release from RE.WO was also higher on average, but the difference did not remain significant after correction for multiple comparisons (BH-adjusted $p = 0.199$). No significant difference was detected under weakly alkaline conditions. These results support the interpretation that the strongest material-dependent contrast concerns the cumulative release of trace elements under acidic conditions, whereas major-element release is more strongly controlled by element-specific mineralogical constraints.

The European fertilising products regulation (2019/1009): constraints and opportunities for BO50 and RE.WO

A comprehensive description of the regulatory framework and criteria is provided in the Supplementary Information SI-2. The regulatory viability of the investigated materials was assessed against the European Fertilising Products Regulation (FPR, Regulation (EU) 2019/1009) and national frameworks (e.g., Italian Legislative Decree No. 75/2010). Under the FPR (SI-2a), CE-marked products must align with specific Product Function Categories (PFCs) and consist solely of approved Component Material Categories (CMCs).

BO50 is a rock powder with a composition similar to other commercial products, often referred to as “rock flour”, currently used as soil improvers, plant biostimulants, or natural fortifiers against biotic and abiotic stresses. The critical distinction lies in the material origin: while commercial powders are specifically produced through grinding, BO50 is a by-product generated during the cutting of the source rock. Paradoxically, this origin may lead to its classification as waste under current regulations, unless it is formally recognized as a by-product under Article 5 of Directive 2008/98/EC (SI-2b, d). Conversely, RE.WO has achieved *End-of-Waste* status but is explicitly excluded from CMC 1 (virgin materials) and does not fit CMC 11 (reserved for non-waste by-products), making its integration into the current CE framework highly challenging (SI-2c, d).

Possible agronomic valorisation strategies for BO50 and RE.WO in the FPR and national frameworks

While the present comparative screening-level assessment defines the intrinsic mineralogical and leaching properties of the materials, it represents a conservative baseline that

does not account for the complex soil-specific controls, such as texture, organic matter, microbial activity, and crop demand, that typically modulate nutrient availability and contaminant mobility in the field.

As discussed above and summarized in Figures S5 and S6, RE.WO generally exhibited higher leaching rates and cumulative releases than BO50 for both major and trace elements, particularly under citric and acetic acid extractions. Conversely, BO50 showed lower overall solubility, largely dominated by K mobilisation, while RE.WO displayed preferential release of Ca, Mg, Fe, and transition metals. Across all samples, citric acid yielded the most extensive extraction, followed by acetic acid, while alkaline water extraction consistently produced the lowest release values. Regarding regulatory compliance, both RE.WO and BO50 contain measurable, potentially problematic concentrations of regulated trace elements, notably Cd and Ni. Depending on the applicable CMC (Annex II of Reg. 2019/1009), these concentrations may slightly exceed the regulatory contaminant thresholds. Cr, on the other hand, is not considered critical as it occurs nearly exclusively in the trivalent form; Cr(VI), the toxic and more mobile species of greatest regulatory concern, was below 1 mg kg^{-1} in both materials according to EN 15192 extraction (Table S1); nevertheless, as a precautionary measure, Cr was not included among the list of possible micronutrients. The same precautionary approach was applied to Ni, Co and Mo based on total content, although the latter two do not present critical concentrations.

Many FPR contaminant limits apply to the product placed on the market; however, as pointed out, the present results should be interpreted as a comparative assessment of element extractability under standardized abiotic conditions, rather than as a direct measure of bioavailability in real soils. Element release was strongly influenced by the extractant: citric acid mobilized the largest fractions of both micronutrients and several trace elements of concern, whereas alkaline extraction consistently yielded the lowest release values (Figs. 3 and 7). Although Cd release remained low in absolute terms, its extractability under chelating conditions was not negligible relative to its bulk concentration, indicating that this element deserves attention in the comparative screening of similar materials. These findings suggest that some elements may become markedly more extractable under abiotic chelating conditions used as an operational proxy for rhizosphere-like processes. This should not be interpreted as direct evidence of biological availability or plant uptake, but rather as a screening-level signal to be evaluated in subsequent soil-based risk assessments and realistic application scenarios using well-characterized soils.

Considering the requirement for analytical verification and risk assessment for each proposed valorisation strategy, several agronomic use options are outlined below, with three



specific product concepts described in Supplementary Information SI-3.

BO50 is a promising mineral soil improver, characterized by high K content, a predominantly crystalline silicate assemblage, and comparatively low concentrations and release of regulated trace elements. It is particularly suitable for acidic or K- and silicate-depleted soils within both integrated and organic farming systems. Practical applications include mixing with compost or organic improvers (5–30 wt. %) to enhance K and micronutrient supply, as well as to improve soil structure and cation exchange capacity over time (see Introduction). Furthermore, BO50 may serve as a functional filler in national (non-CE) organo-mineral fertilizers, replacing inert components (SI-3a). Similarly, it could partially substitute virgin rock flours (e.g., feldspars, porphyry, basalts) in siliceous materials, typically within the range of 20–50 wt. %. Alternatively, to secure formal non-waste status under EU regulations (SI-2b,d), BO50 could be integrated into EU fertilising products as a component material (CMC 11) for soil improvers or liming materials, provided it demonstrates significant neutralizing capacity for the latter. Implementation of this proposal requires REACH registration, explicitly covering its use as a fertilising material within the Chemical Safety Report.

RE.WO exhibits faster release kinetics than BO50, suggesting potential for applications requiring relatively rapid nutrient delivery under abiotic acidic/chelating conditions. However, this advantage is accompanied by concentrations of certain regulated trace elements that exceed regulatory thresholds. The variability in these levels may partly reflect sampling limitations, as RE.WO is still an experimental material with a limited production scale. Pending clearer EU-level classification pathways, any potential soil-related use of RE.WO appears more defensible in restricted non-food applications, such as ornamental lawns, urban green spaces, or industrial sites (SI-3b, c). In these contexts, materials may occasionally be evaluated through site-specific, risk-based approaches. Alternatively, RE.WO could be considered at low application rates in formulations with clean materials (e.g., peat, selected composts, sands), provided that the final product is justified by a specific functional contribution of RE.WO, such as structural porosity, drainage behaviour, or micronutrient release, rather than by the mere reduction of regulated trace metal concentrations through dilution. In such cases, metal concentrations in the final product, cumulative soil loading, and compliance with national regulatory limits must be carefully assessed. Given its high Zn and Cu content (and Cr, conservatively excluded here), RE.WO may also have potential as a micronutrient source, although this interpretation remains provisional. Furthermore, RE.WO may serve as a structural component in growing media for container cultivation (e.g., vertical farming), where porosity and drainage are critical. In this respect,

it differs from conventional rock wool as a non-fibrous mineral material obtained through thermal processing.

However, even in this scenario, compliance with the permitted CMCs for PFC 4 (Growing Media) must be assessed, as RE.WO does not clearly fit within any existing CMC (SI-2c, d). Beyond CMC eligibility, repeated application also requires a cumulative-risk perspective: the comparatively high total Ni content of RE.WO, together with measurable Cd release under chelating conditions, indicates that repeated soil applications may raise cumulative loading concerns. Accordingly, any direct repeated use of RE.WO in food-producing agricultural soils should remain conditional on cumulative loading assessment, batch-level quality control and soil-based validation.

To further contextualize the possible valorisation strategies, the 72h cumulative release data were converted into apparent element release per unit surface area for selected application-rate scenarios (Table S2). The calculation was performed as a simple mass-balance conversion using the citric-acid extraction, which produced the highest overall element release among the organic-acid conditions tested and served as an abiotic proxy for chelating rhizosphere-like processes. These estimates should not be interpreted as application recommendations or as predictions of plant-available fluxes because, as mentioned, they do not account for soil retention, secondary precipitation, biological uptake, leaching losses, or long-term weathering. Nevertheless, they provide useful screening-level context for comparing the two materials. At $t \text{ ha}^{-1}$ -scale application rates, BO50 would mainly contribute through K release, with comparatively low apparent release of regulated trace elements under the tested conditions. In contrast, RE.WO would release larger amounts of Mn, Cu and Zn per unit mass, but also higher Ni and measurable Cd, indicating that repeated use should be evaluated through cumulative loading calculations and soil-based validation. Accordingly, application scenarios for RE.WO appear more defensible at low dose, in blended formulations, or in restricted non-food/technical substrates until soil incubation, column and plant-growth experiments in well-characterized soils are available.

Conclusion

Overall, this comparative abiotic assessment indicates that short-term nutrient release and trace-metal mobility are closely linked to the mineralogical contrast between the glassy matrix of RE.WO and the predominantly crystalline silicate assemblage of BO50. RE.WO shows substantially higher solubility and leaching efficiency for both major and trace elements than BO50. Its amorphous glassy structure favours rapid nutrient release but also enhances the mobilization of regulated trace metals. In contrast, BO50,



dominated by crystalline silicates, exhibits a much lower leaching potential, making it less effective as a micronutrient supplier but intrinsically more stable in terms of trace-metal release. BO50 and RE.WO should therefore be regarded as candidate component materials for specific soil-related applications, rather than as validated agronomic products. Soil incubation, column leaching (including pH monitoring), and plant-growth experiments are required to determine actual nutrient bioavailability and to verify how the observed release patterns evolve under realistic soil conditions, where sorption, microbial activity and mineral transformation may influence biological compatibility and modulate both nutrient availability and trace-metal mobility.

The evaluation of BO50 and RE.WO within the framework of Regulation (EU) 2019/1009 highlights both the regulatory constraints and the potential opportunities for their agronomic valorisation. BO50, whose composition closely resembles commercially available rock flours, could be used as a soil-improving mineral input under national (non-CE) frameworks and, if formally recognised as a by-product and fully compliant with REACH, may ultimately qualify for inclusion under CMC 11. RE.WO, despite exhibiting favourable release properties, still faces regulatory uncertainty for direct soil-related use, particularly in view of its regulated trace metal contents and the current lack of clear EU pathways for recovered mineral materials; accordingly, any potential use appears more defensible in restricted non-food applications or at low application rates, subject to case-specific risk assessment.

The proposed concepts (SI-3) demonstrate that both materials can contribute to circular practices: BO50 as an effective siliceous-potassic soil improver, and RE.WO as either a component of technical substrates or a minor additive for micronutrient release where risk is manageable. Nevertheless, when materials analogous to BO50 are traded for alternative uses (e.g., rock powder for foliar application as bio-stimulants), regulatory requirements may become less stringent. While this does not inherently exempt the materials from the FPR, it shifts the regulatory framework, prompting a critical discussion regarding claimed functionality and strategic market placement.

BO50 and RE.WO may also offer agronomic advantages beyond their regulatory positioning, although, as mentioned above, confirmation under field conditions is needed. Their gradual and sustained nutrient release suggests potential to support more sustainable nutrient management by reducing reliance on synthetic fertilisers and lowering the frequency of external inputs. In contexts where conventional soil improvers are costly or difficult to access, BO50 and RE.WO may contribute to long-term soil fertility while complementing integrated soil management strategies. In addition to nutrient supply, their mineralogical composition may provide ancillary benefits, such as progressive buffering

of soil acidity, as reported for comparable silicate-rich rocks like virgin basalt.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13762-026-07435-3>.

Acknowledgement We thank Zetadi s.r.l. (Ferno, Varese, Italy) (<https://www.zetadisrl.it/>) and SIBELCO (<https://www.sibelco.com/>) for providing RE.WO and BO50 materials, respectively.

Funding Open access funding provided by Università degli Studi di Modena e Reggio Emilia within the CRUI-CARE Agreement. This work was funded under: i) PRIN2022 “SEEDS—Sediments Eco-recycling Exploitation, Development and Sustainability”—Project Code 2022BCL34N, Finanziato dall’Unione Europea- Next Generation EU, Missione 4 Componente 1 CUP-E53D23004330006; ii) Project 101113855 LIFE22-ENV-IT-LIFE: RE.WO, REcycling mineral WOol waste into high-value products. iii) ECOSISTER: National Recovery and Resilience Plan (NRRP), Mission 04 Component 2 Investment 1.5—Next GenerationEU, Call for tender n. 3277 dated 30/12/2021; Award Number: 0001052 dated 23/06/2022.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Abdul Halim NS, Abdullah R, Karsani SA et al (2018) Influence of soil amendments on the growth and yield of rice in acidic soil. *Agronomy* (Basel) 8:165. <https://doi.org/10.3390/agronomy8090165>
- Anda M, Shamshuddin J, Fauziah CI (2015) Improving chemical properties of a highly weathered soil using finely ground basalt rocks. *CATENA* 124:147–161. <https://doi.org/10.1016/j.catena.2014.09.012>
- Andrade JFC, Córdoba RE, Schalch V (2025) A review of leaching testing methods and identification of key pollutants in leachate from construction and demolition waste. *Int J Environ Sci Technol* 22:9637–9654. <https://doi.org/10.1007/s13762-025-06474-6>
- Arletti R, Conte S, Zanelli C et al (2023) Recycling the product of thermal inertization of man-made vitreous fibres for the manufacture of stoneware tiles. *Waste Biomass Valor* 14:1721–1736. <https://doi.org/10.1007/s12649-022-01959-8>



- Aslan MU, Arslan H (2024) The effect of zeolites on soil and plant: a review. *Commun Soil Sci Plant Anal* 55:2197–2216. <https://doi.org/10.1080/00103624.2024.2346211>
- Benevides Filho PRR, Blaskowski AE, Ramos MN et al (2023) Potential soil remineralizers from silicate rock powders (SRP) as alternative sources of nutrients for agricultural production (Amazon Region). *Minerals* 13:1255. <https://doi.org/10.3390/min13101255>
- Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc Ser B Stat Methodol* 57:289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Blake RL, Hesseck RE, Zoltai T, Finger LW (1966) Refinement of the hematite structure. *Am Mineral* 51:123–129
- Borah D, Nath H, Saikia H (2022) Modification of bentonite clay & its applications: a review. *Rev Inorg Chem* 42:265–282. <https://doi.org/10.1515/revic-2021-0030>
- Botsou F, Sungur A, Kelepertzis E, Soyak M (2016) Insights into the chemical partitioning of trace metals in roadside and off-road agricultural soils along two major highways in Attica's region, Greece. *Ecotoxicol Environ Saf* 132:101–110. <https://doi.org/10.1016/j.ecoenv.2016.05.032>
- Brigatti MF, Guidotti CV, Malferrari D, Sassi FP (2008) Single-crystal X-ray studies of trioctahedral micas coexisting with dioctahedral micas in metamorphic sequences from western Maine. *Am Mineral* 93:396–408. <https://doi.org/10.2138/am.2008.2523>
- Cataldo E, Salvi L, Paoli F et al (2021) Application of zeolites in agriculture and other potential uses: a review. *Agronomy* 11(8):1547. <https://doi.org/10.3390/agronomy11081547>
- Chen H, Su X, Luo Y et al (2024) Natural-derived porous nanocarriers for the delivery of essential oils. *Chin J Nat Med* 22:1117–1133. [https://doi.org/10.1016/S1875-5364\(24\)60731-4](https://doi.org/10.1016/S1875-5364(24)60731-4)
- Chiglino L, Ballesterio J, Celio A et al (2022) Basalts of the Arapey group and their potential use as a source of agrominerals (stone-meal). *Agrociencia Urug* 26:A505. <https://doi.org/10.31285/AGRO.26.505>
- Chokri M, Azougagh O, El Bojaddayni I et al (2025) Progress in bentonite clay modification and enhancing properties to industrial applications: a review. *Mater Chem Phys* 337:130486. <https://doi.org/10.1016/j.matchemphys.2025.130486>
- Conceição LT, Silva GN, Holsback HMS et al (2022) Potential of basalt dust to improve soil fertility and crop nutrition. *J Agric Food Res* 10:100443. <https://doi.org/10.1016/j.jafr.2022.100443>
- Dalmora AC, Ramos CG, Querol X et al (2016) Nanoparticulate mineral matter from basalt dust wastes. *Chemosphere* 144:2013–2017. <https://doi.org/10.1016/j.chemosphere.2015.10.047>
- Datta R, Holatko J, Latal O et al (2020) Bentonite-based organic amendment enriches microbial activity in agricultural soils. *Land* 9:258. <https://doi.org/10.3390/land9080258>
- De Carvalho Ribeiro M, Redondo Martins A, Silva Alves R et al (2025) Basalt rock dust as a soil remineralizer in upland rice grown in tropical soils: residual effects on soil fertility, Si availability, and leaf anatomy. *J Clean Prod* 520:146063. <https://doi.org/10.1016/j.jclepro.2025.146063>
- De Carvalho Ribeiro M, Ganga A, Cattanio IS et al (2024) Residual effect of silicate agromineral application on soil acidity, mineral availability, and soybean anatomy. *Agronomy* 15:5. <https://doi.org/10.3390/agronomy15010005>
- De Smedt C, Someus E, Spanoghe P (2015) Potential and actual uses of zeolites in crop protection. *Pest Manag Sci* 71:1355–1367. <https://doi.org/10.1002/ps.3999>
- Desmalles C, Jordan-Meille L, Hernandez J et al (2025) Impact of basalt rock powder on ryegrass growth and nutrition on sandy and loamy acid soils. *Agronomy* 15:1791. <https://doi.org/10.3390/agronomy15081791>
- DGL75–2010 (2010) Riordino e revisione della disciplina in materia di fertilizzanti, a norma dell'articolo 13 della legge 7 luglio 2009, n. 88.
- Ding YZ, Song ZG, Feng RW, Guo JK (2014) Interaction of organic acids and pH on multi-heavy metal extraction from alkaline and acid mine soils. *Int J Environ Sci Technol* 11:33–42. <https://doi.org/10.1007/s13762-013-0433-7>
- Dupla X, Claustre R, Bonvin E et al (2024) Let the dust settle: impact of enhanced rock weathering on soil biological, physical, and geochemical fertility. *Sci Total Environ* 954:176297. <https://doi.org/10.1016/j.scitotenv.2024.176297>
- Eslami M, Khorassani R, Coltorti M et al (2018) Leaching behaviour of a sandy soil amended with natural and NH₄⁺ and K⁺ saturated clinoptilolite and chabazite. *Arch Agron Soil Sci* 64:1142–1151. <https://doi.org/10.1080/03650340.2017.1414944>
- EU 2019/1009 (2019) Regulation (EU) 2019/1009 of the European Parliament and of the Council of 5 June 2019 laying down rules on the making available on the market of EU fertilising products and amending Regulations (EC) No 1069/2009 and (EC) No 1107/2009 and repealing Regulation (EC) No 2003/2003
- Faccini B, Di Giuseppe D, Malferrari D et al (2015) Ammonium-exchanged zeolite preparation for agricultural uses: From laboratory tests to large-scale application in ZeoLIFE project prototype. *Period Miner* 84:303–321. <https://doi.org/10.2451/2015PM0015>
- Galardini G, Ferretti G, Medoro V et al (2024a) Applying natural and K-enriched zeolite before struvite precipitation improved the recovery of NH₄⁺ from liquid digestate and the reagent use efficiency. *Int J Environ Res* 18:44. <https://doi.org/10.1007/s41742-024-00595-5>
- Galardini G, Malferrari D, Altimari F et al (2024b) From quarry by-products to a zeolites-based Zn fertilizer with increased resistance to rain leaching. *Microporous Mesoporous Mater* 379:113290. <https://doi.org/10.1016/j.micromeso.2024.113290>
- Glenn DM, Puterka G (2004) Particle film technology: an overview of history, concepts and impact in horticulture. *Acta Hort* 636:509–511. <https://doi.org/10.17660/ActaHortic.2004.636.63>
- Goyal A, Chavan SS, Mohite RA et al (2025) Emerging trends and perspectives on nano-fertilizers for sustainable agriculture. *Discover Nano* 20:97. <https://doi.org/10.1186/s11671-025-04286-8>
- Gualtieri AF (2000) Accuracy of XRPD QPA using the combined Rietveld–RIR method. *J Appl Crystallogr* 33:267–278. <https://doi.org/10.1107/S002188989901643X>
- Gualtieri AF, Zanatto I (2022) Apparatus for treating waste containing mineral wool. *Eur Pat N* 22153845:7
- Harlow GE (1982) The anorthoclase structures: the effects of temperature and composition. *Am Miner* 67:975–996
- Hartatik W, Wibowo H, Anda M (2019) Potential of mineral resources as an alternative raw material for K fertilizer. *IOP Conf Ser Earth Environ Sci* 393:012085. <https://doi.org/10.1088/1755-1315/393/1/012085>
- Jarosz R, Szerement J, Gondek K, Mierzwa-Hersztek M (2022) The use of zeolites as an addition to fertilisers – a review. *CATENA* 213:106125. <https://doi.org/10.1016/j.catena.2022.106125>
- Jones DL (1998) Organic acids in the rhizosphere – a critical review. *Plant Soil* 205:25–44. <https://doi.org/10.1023/A:1004356007312>
- Khademi Z, Jones DL, Malakouti MJ et al (2009) Organic acid mediated nutrient extraction efficiency in three calcareous soils. *Soil Res* 47:213. <https://doi.org/10.1071/SR07179>
- Son NK, Khoi NHT, Huynh NNT, Phuoc NV (2024) comprehensive review of rock dust for soil remineralization in sustainable agriculture and preliminary assessment of nutrient values in micronized porous basalt rock from Nghe-An province Vietnam. *Vietnam J Sci Technol* 63(5):947–959. <https://doi.org/10.15625/2525-2518/18544>



- Korchagin J, Caner L, Bortoluzzi EC (2019) Variability of amethyst mining waste: a mineralogical and geochemical approach to evaluate the potential use in agriculture. *J Clean Prod* 210:749–758. <https://doi.org/10.1016/j.jclepro.2018.11.039>
- Korchagin J, Escosteguy PAV, Bortoluzzi EC (2022) Nutrient transfer in rangelands under rock powder amendment. *J Plant Nutr Soil Sci* 185:656–667. <https://doi.org/10.1002/jpln.202200059>
- Le Page Y, Donnay G (1976) Refinement of the crystal structure of low-quartz. *Acta Crystallogr B* 32:2456–2459. <https://doi.org/10.1107/S0567740876007966>
- Lewis AL, Sarkar B, Wade P et al (2021) Effects of mineralogy, chemistry and physical properties of basalts on carbon capture potential and plant-nutrient element release via enhanced weathering. *Appl Geochem* 132:105023. <https://doi.org/10.1016/j.apgeochem.2021.105023>
- Lu P, Apps J, Zhang G et al (2024) Knowledge gaps and research needs for modeling CO₂ mineralization in the basalt-CO₂-water system: a review of laboratory experiments. *Earth-Sci Rev* 254:104813. <https://doi.org/10.1016/j.earscirev.2024.104813>
- Luchese AV, De Castro Leite IJG, Alves ML et al (2023a) Can basalt rock powder be used as an alternative nutrient source for soybeans and corn? *J Soil Sci Plant Nutr* 23:4044–4054. <https://doi.org/10.1007/s42729-023-01322-3>
- Luchese AV, Gutz De Castro Leite II, Da Silva Giarretta AP et al (2023b) Use of quarry waste basalt rock powder as a soil remineralizer to grow soybean and maize. *Heliyon* 9:e14050. <https://doi.org/10.1016/j.heliyon.2023.e14050>
- Malferrari D, Laurora A, Brigatti MF et al (2021) A time- and cost-saving method to check the point-to-point distribution of soil improvers. *J Plant Nutr Soil Sci* 184:263–267. <https://doi.org/10.1002/jpln.202000409>
- Mancini F, Sillanpää R, Marshall B, Papunen H (1996) Magnesian hornblende from a metamorphosed ultramafic body in Southwest Finland; crystal chemistry and petrological implications. *Can Mineral* 34:835–844
- Mi J, Gregorich EG, Xu S et al (2020) Effect of bentonite as a soil amendment on field water-holding capacity, and millet photosynthesis and grain quality. *Sci Rep* 10:18282. <https://doi.org/10.1038/s41598-020-75350-9>
- Nakachew K, Yigermal H, Assefa F et al (2024) Review on enhancing the efficiency of fertilizer utilization: strategies for optimal nutrient management. *Open Agric* 9:20220356. <https://doi.org/10.1515/opag-2022-0356>
- Nunes JMG, Kautzmann RM, Oliveira C (2014) Evaluation of the natural fertilizing potential of basalt dust wastes from the mining district of Nova Prata (Brazil). *J Clean Prod* 84:649–656. <https://doi.org/10.1016/j.jclepro.2014.04.032>
- Palmer DC, Dove MT, Ibberson RM, Powell BM (1997) Structural behavior, crystal chemistry, and phase transitions in substituted leucite; high-resolution neutron powder diffraction studies. *Am Mineral* 82:16–29. <https://doi.org/10.2138/am-1997-1-203>
- Petoumenou DG (2023) Enhancing yield and physiological performance by foliar applications of chemically inert mineral particles in a rainfed vineyard under Mediterranean conditions. *Plants (Basel)* 12:1444. <https://doi.org/10.3390/plants12071444>
- Quille P, Kacprzyk J, O'Connell S, Ng CKY (2025) Reducing fertiliser inputs: plant biostimulants as an emerging strategy to improve nutrient use efficiency. *Discov Sustain* 6:128. <https://doi.org/10.1007/s43621-025-00910-w>
- Ramos CG, Querol X, Oliveira MLS et al (2015) A preliminary evaluation of volcanic rock powder for application in agriculture as soil a remineralizer. *Sci Total Environ* 512–513:371–380. <https://doi.org/10.1016/j.scitotenv.2014.12.070>
- Ramos CG, Querol X, Dalmora AC et al (2017) Evaluation of the potential of volcanic rock waste from southern Brazil as a natural soil fertilizer. *J Clean Prod* 142:2700–2706. <https://doi.org/10.1016/j.jclepro.2016.11.006>
- Reddy MK, Asthana R, Debnath S et al (2024) Nanofertilizers for sustainable crop production: a comprehensive review. *BioNanoScience* 14:1918–1939. <https://doi.org/10.1007/s12668-024-01413-0>
- Rodrigues M, Bortolini PC, Neto CK et al (2024) Unlocking higher yields in *Urochloa brizantha*: the role of basalt powder in enhancing soil nutrient availability. *Discov Soil* 1:4. <https://doi.org/10.1007/s44378-024-00006-3>
- Rodrigues M, Neto CK, Passos AID et al (2025) Sustainable soil amendment with basalt powder: unveiling integrated soil-plant responses in *Ilex paraguariensis* cultivation. *AgriEngineering* 7:290. <https://doi.org/10.3390/agriengineering7090290>
- Rotondi A, Bertazza G, Faccini B et al (2022) Effect of different foliar particle films (kaolin and zeolitite) on chemical and sensory properties of olive oil. *Agronomy* 12:3088. <https://doi.org/10.3390/agronomy12123088>
- Sah SK, Reddy KR, Li J (2022) Silicon enhances plant vegetative growth and soil water retention of soybean (*Glycine max*) plants under water-limiting conditions. *Plants* 11:1687. <https://doi.org/10.3390/plants11131687>
- Salerno G, Rebora M, Piersanti S et al (2021) Reduction in insect attachment caused by different nanomaterials used as particle films (Kaolin, Zeolite, Calcium Carbonate). *Sustainability* 13:8250. <https://doi.org/10.3390/su13158250>
- Sarkar S, Upadhyay PK, Mitran T et al (2025) The multifaceted role of zeolites in modern agriculture and environmental management. *J Plant Nutr*. <https://doi.org/10.1080/01904167.2025.2518245>
- Sharma H, Sharma N (2025) A review on the efficient release of nutrients for sustainable agriculture. *J Soil Sci Plant Nutr* 25:7061–7073. <https://doi.org/10.1007/s42729-025-02580-z>
- Sharma RR, Vijay Rakesh Reddy S, Datta SC (2015) Particle films and their applications in horticultural crops. *Appl Clay Sci* 116–117:54–68. <https://doi.org/10.1016/j.clay.2015.08.009>
- Sungur A, Soylak M, Yilmaz E et al (2015) Characterization of heavy metal fractions in agricultural soils by sequential extraction procedure: the relationship between soil properties and heavy metal fractions. *Soil Sediment Contam Int J* 24:1–15. <https://doi.org/10.1080/15320383.2014.907238>
- Sur IM, Hegyi A, Micle V et al (2023) Influence of the extraction solution on the removal of heavy metals from polluted soils. *Materials* 16:6189. <https://doi.org/10.3390/ma16186189>
- Swoboda P, Döring TF, Hamer M (2022) Remineralizing soils? The agricultural usage of silicate rock powders: a review. *Sci Total Environ* 807:150976. <https://doi.org/10.1016/j.scitotenv.2021.150976>
- Väntsi O, Kärki T (2014) Mineral wool waste in Europe: a review of mineral wool waste quantity, quality, and current recycling methods. *J Mater Cycles Waste Manag* 16:62–72. <https://doi.org/10.1007/s10163-013-0170-5>
- Violante A, Cozzolino V, Perelomov L et al (2010) Mobility and bio-availability of heavy metals and metalloids in soil environments. *J Soil Sci Plant Nutr* 10:268–292. <https://doi.org/10.4067/S0718-95162010000100005>
- Yamnova NA, Khomyakov AP, Zlykhenskaya IV (2000) Refinement of the crystal structure of sanidine-like feldspar. *Crystallogr Rep* 45:754–758. <https://doi.org/10.1134/1.1312914>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

